

Social Robots in Therapy and Care

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Abstract—The Social Robots in Therapy workshop series aims at advancing research topics related to the use of robots in the contexts of Social Care and Robot-Assisted Therapy (RAT). Robots in social care and therapy have been a long time promise in HRI as they have the opportunity to improve patients life significantly. Multiple challenges have to be addressed for this, such as building platforms that work in proximity with patients, therapists and health-care professionals; understanding user needs; developing adaptive and autonomous robot interactions; and addressing ethical questions regarding the use of robots with a vulnerable population.

The full-day workshop follows last year's edition which centered on how social robots can improve health-care interventions, how increasing the degree of autonomy of the robots might affect therapies, and how to overcome the ethical challenges inherent to the use of robot assisted technologies. This 2nd edition of the workshop will be focused on the importance of equipping social robots with socio-emotional intelligence and the ability to perform meaningful and personalized interactions.

This workshop aims to bring together researchers and industry experts in the fields of Human-Robot Interaction, Machine Learning and Robots in Health and Social Care. It will be an opportunity for all to share and discuss ideas, strategies and findings to guide the design and development of robot-assisted systems for therapy and social care implementations that can provide personalize, natural, engaging and autonomous interactions with patients (and health-care providers).

Index Terms—Social Robots, Robots in Therapy; Social Intelligence; Supervised Autonomy; Personalized Behaviors; Adaptive Behaviors.

INTRODUCTION

An increasing number of projects are being developed employing social robots for therapeutic assistance in health-care, with the overall ambitious goal of increasing the well-being of a vulnerable populations; from children to adults, with particular attention, for elderly care, individuals on the autism spectrum, and physical rehabilitation. Robot-Assisted Therapy (RAT) has successfully been used in Human Robot Interaction (HRI) research by including social robots in health-care interventions by virtue of their ability to engage human users in both social and emotional dimensions.

Robots used for these tasks must be designed with several user groups in mind, including both individuals receiving therapy and care professionals responsible for the treatment. These robots must also be able to perceive their context of use, recognize human actions and intentions, and follow the therapeutic goals to perform meaningful and personalized treatment. Effective interactions require for robots to be capable of coordinated, timely behavior in response to social cues. This means being able to estimate and predict levels of engagement, attention, intentionality and emotional state during human-robot interactions. An additional challenge for social robots in therapy and care is the wide range of needs and conditions the different users can have during their interventions, even if they may share the same pathologies their current requirements and the objectives of their therapies can varied extensively. Therefore, it becomes crucial for robots to adapt their behaviors and interaction scenario to the specific needs, preferences and requirements of the patients they interact with. This personalization should be considered in terms of the robot behavior and the intervention scenario and must reflect the needs, preferences and requirements of the user.

The second iteration of this workshop builds on the success of the previous version hosted at the *13th ACM/IEEE International Conference on Human-Robot Interaction*, we aim to discuss the approaches and challenges of developing these social robots and discuss recent advances and research contributions on HRI in the RAT and social robot care contexts on the following challenges: How to conduct long-term studies with robot-assisted therapy *in the wild*? How can long-term, personalized and persistent relationships between robot and user be established, and what does such relationships comprise? What requirements control architectures of social robots should acquire for autonomous behavior in health-care interventions? What are the appropriate level of autonomy for robot in therapy and care? How will robots autonomously read and interpret social signals during interactions? How to generate models of social interactions allowing the robot to adapt to the user and follow the rules and goals of the therapy?

OUTLINE OF THE WORKSHOP

In this workshop we want to gather, compare, and combine knowledge gained in various HRI projects around the world with social robots in therapy and for social care applications. It is the aim of this workshop to reflect on the types of robots used in RAT and ongoing research on Social Robots in the domestic and care facility contexts with a focus on personalized, engaging and adaptable HRI.

A. Format and Activities

This is a full-day workshop organized in thematic sessions. Sessions are composed of invited talks along with participants' short position paper presentations. At the end of the workshop the organizers will initiate semi-structured group discussions to stimulate discussion around the key topics of the workshop, identified by the organizers based on the submissions.

B. List of Topics

The following topics should be addressed in the workshop and guide the submission of papers:

- Social intelligence for robots in therapy and care.
- Adaptive mechanisms for robot used in therapy and care.
- Personalize and persistent HRI in therapy and care.
- Affect and engagement in therapy and care HRI.
- Trust and collaboration in therapy and care HRI.
- Modeling patients, therapist, and robots relationships.
- Shared or full autonomy in robots in therapy and care.
- User and patient-centered HRI studies.
- Long-term therapeutic interaction studies.
- Acceptance studies on social robots used in therapy.
- Clinical validation of robot-assisted therapies.
- Ethical challenges for social care robots.

C. Invited Speakers

- Tom Ziemke - University of Skövde and Linköping University, Sweden.
- Cindy Bethel - Mississippi State University, USA.
- Ognjen Rudovic - MIT Media Lab, USA.
- Yukie Nagai - NICT, Japan.
- Takayuki Kanda - Kyoto University, Japan

TARGET AUDIENCE

The primary audience of the workshop are researchers in the field of Human Robot Interaction, Robot-Assisted Therapy and Robots in Health and Social Care. In particular, researchers working with the elderly, individuals with physical or cognitive impairments and in rehabilitation therapy, patient-centered HRI evaluation, the ethical and moral implications of robotics in health and social care will be interested in this workshop.

The use of robots in social therapies requires a highly interdisciplinary collaboration among engineers, roboticist, computer scientists, psychologist, therapists, ethicists health-care professionals. In general, we are convinced that this workshop will attract a broad target audience, as the topic is of huge interest in the community. We expect an interdisciplinary audience.

ORGANIZERS

Daniel Hernández García is a postdoctoral research fellow in Human-Robot Interaction at the Centre for Robotics and Neural Systems in the University of Plymouth (United Kingdom). He received his Ph.D. degree in Electrical Engineering, Electronics and Automation from the Universidad Carlos III of Madrid. His current research focuses on HRI in social robots for education and therapy. He co-organizer of the first edition of this workshop.

Pablo Gómez Esteban is a postdoctoral research fellow in HRI at the Robotics and Multibody research group at the Vrije Universiteit Brussel (Belgium). He holds a PhD in decision support systems from the University Rey Juan Carlos. His current research focuses on affective computing and social robots in therapy for Human-Robot Interaction. He was co-organizer of the first edition of this workshop.

Hee Rin Lee is a postdoctoral scholar of Computer Science and Engineering at UC San Diego. She earned her PhD in Informatics at Indiana University Bloomington. She is interested in co-design methodologies to incorporate older adults perspectives into assistive robot design processes so that the robot designs can properly reflect their issues and concerns. She was co-organizer of the first edition of this workshop.

Marta Romeo is currently completing her PhD as part of the Machine Learning and Optimization group at the University of Manchester (United Kingdom) under the supervision of Prof. Angelo Cangelosi. She graduated with a bachelor and a master in Computer Engineering from Politecnico di Milano, with a particular focus on Artificial Intelligence and Robotics. Her current research focuses on machine learning for HRI and elderly care.

Emmanuel Senft is currently completing his PhD at the University of Plymouth (United Kingdom) under the supervision of Prof. Tony Belpaeme. After graduating from EPFL with a Bachelor and a master in Robotic and Autonomous Systems, he moved to Plymouth to research how robot can be taught by humans to interact socially in our society. His approach, SPARC, allows humans to teach intuitively and quickly behaviors to robots while ensuring that the robots behaviors are always safe for the interaction partners.

Erik Billing is a Senior Lecturer at the University of Skövde (Sweden). With a master in cognitive science and a PhD in computing science, he is engaged with research on human-robot interaction, user experience, and robot learning from human demonstration. He has done work on cognitive modeling of both human and animal behavior and is currently building up the world's first master program on Human Robot Interaction, hosted at the University of Skövde.

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